AI in Orthopedic Research: A Comprehensive Review

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CLINICAL & AGENTIC AI TAKEAWAY

AI is revolutionizing orthopedic research and clinical practice by improving diagnostic accuracy, optimizing treatment strategies, and streamlining clinical workflows, but challenges like data heterogeneity and algorithmic bias remain.

ABSTRACT

Artificial intelligence (AI) is revolutionizing orthopedic research and clinical practice by enhancing diagnostic accuracy, optimizing treatment strategies, and streamlining clinical workflows. Recent advances in deep learning have enabled the development of algorithms that detect fractures, grade osteoarthritis, and identify subtle pathologies in radiographic and magnetic resonance images with performance comparable to expert clinicians. These AI-driven systems reduce missed diagnoses and provide objective, reproducible assessments that facilitate early intervention and personalized treatment planning. Moreover, AI has made significant strides in predictive analytics by integrating diverse patient data-including gait and imaging features-to forecast surgical outcomes, implant survivorship, and rehabilitation trajectories. Emerging applications in robotics, augmented reality, digital twin technologies, and exoskeleton control promise to further transform preoperative planning and intraoperative guidance. Despite these promising developments, challenges such as data heterogeneity, algorithmic bias, and the "black box" nature of many models-as well as issues with robust validation-remain. This comprehensive review synthesizes current developments, critically examines limitations, and outlines future directions for integrating AI into musculoskeletal care. © 2025 Orthopaedic Research Society.

Keywords: Cartilage, Biomechanics, Rehabilitation, Osteoarthritis, Diagnostic Imaging, Disease Process, Surgical Navigation Outcomes

Study Overview & Background

Artificial intelligence (AI) is revolutionizing orthopedic research and clinical practice by enhancing diagnostic accuracy, optimizing treatment strategies, and streamlining clinical workflows. Recent advances in deep learning have enabled the development of algorithms that detect fractures, grade osteoarthritis, and identify subtle pathologies in radiographic and magnetic resonance images with performance comparable to expert clinicians. These AI-driven systems reduce missed diagnoses and provide objective, reproducible assessments that facilitate early intervention and personalized treatment planning. Moreover, AI has made significant strides in predictive analytics by integrating diverse patient data-including gait and imaging features-to forecast surgical outcomes, implant survivorship, and rehabilitation trajectories. Emerging applications in robotics, augmented reality, digital twin technologies, and exoskeleton control promise to further transform preoperative planning and intraoperative guidance. Despite these promising developments, challenges such as data

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Methodology & Clinical Focus

Study Classification: literature review. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

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Extended Repository Abstract

Artificial intelligence (AI) is revolutionizing orthopedic research and clinical practice by enhancing diagnostic accuracy, optimizing treatment strategies, and streamlining clinical workflows. Recent advances in deep learning have enabled the development of algorithms that detect fractures, grade osteoarthritis, and identify subtle pathologies in radiographic and magnetic resonance images with performance comparable to expert clinicians. These AI-driven systems reduce missed diagnoses and provide objective, reproducible assessments that facilitate early intervention and personalized treatment planning. Moreover, AI has made significant strides in predictive analytics by integrating diverse patient data-including gait and imaging features-to forecast surgical outcomes, implant survivorship, and rehabilitation trajectories. Emerging applications in robotics, augmented reality, digital twin technologies, and exoskeleton control promise to further transform preoperative planning and intraoperative guidance. Despite these promising developments, challenges such as data heterogeneity, algorithmic bias, and the "black box" nature of many models-as well as issues with robust validation-remain. This comprehensive review synthesizes current developments, critically examines limitations, and outlines future directions for integrating AI into musculoskeletal care.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

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